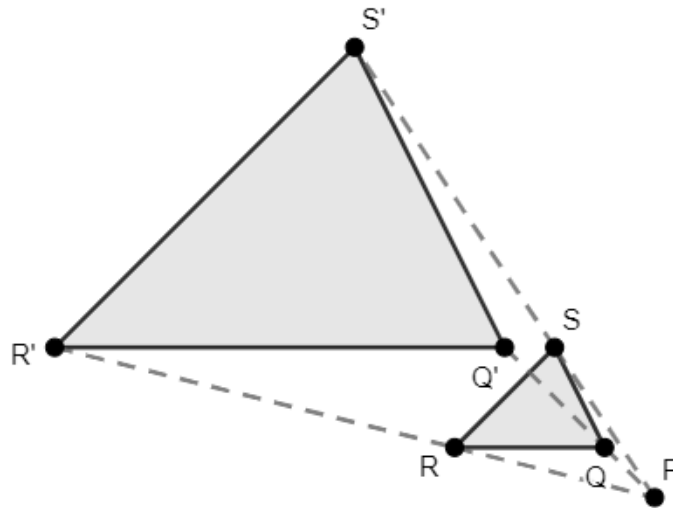


$\triangle Q'R'S'$ was created by dilating $\triangle QRS$. The length of $\overline{QR}$ is 3 and the length of $\overline{Q'R'}$ is 9.



1. $\triangle QRS$ is the ☐ image ☒ preimage, $\triangle Q'R'S'$ is the ☒ image ☐ preimage, and the center of

dilation is point ☒ P ☐ Q ☐ Q'.

2. Determine the dilation ratio.

$$\frac{Q'R'}{QR} = \frac{9}{3} = 3$$

3. If the length of $\overline{Q'S'}$ is $3\sqrt{5}$, determine the length of $\overline{QS}$.
 $QS = \sqrt{5}$

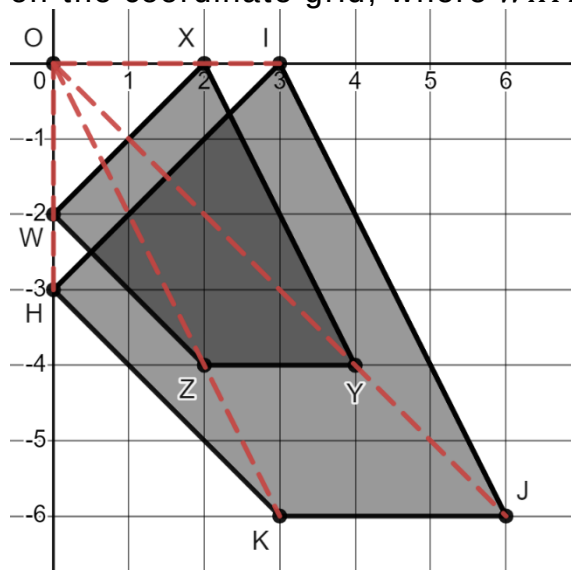
4. What is the ratio of $\frac{RS}{R'S'}$?

$$\frac{RS}{R'S'} = \frac{1}{3}$$

5. What is the ratio of $\frac{P'R'}{PR}$?

$$\frac{P'R'}{PR} = \frac{3}{1} = 3$$

WXYZ and HIJK are shown on the coordinate grid, where WXYZ is a dilation of HIJK.



6. Complete the table.

Segment	Slope	Distance
$\overline{YZ}$	0	2
$\overline{JK}$	0	3
$\overline{WX}$	$\frac{2}{2} = 1$	$\sqrt{8} = 2\sqrt{2}$
$\overline{HI}$	$\frac{3}{3} = 1$	$\sqrt{18} = 3\sqrt{2}$

7. Show the dilation on the coordinate grid by drawing dotted lines from the center of dilation through each vertex.

8. Write a description of the dilation by giving the center of dilation and the scale factor.

$(x,y) \rightarrow (\frac{2}{3}x, \frac{2}{3}y)$
Center of dilation is (0,0) and scale factor is $\frac{2}{3}$

9. What is the ratio $\frac{OX}{OI}$?

$\frac{OX}{OI} = \frac{2}{3}$

10. What is the ratio $\frac{OX}{XI}$?

$\frac{OX}{XI} = \frac{2}{1}$